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Agronomic and economic effects of wetlands on crop yields using precision agriculture data

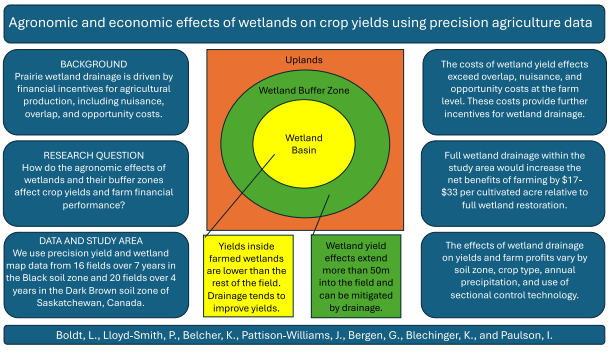
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HIGHLIGHTS

- The agronomic effects of wetlands have rarely been considered in previous studies of wetland conservation costs in the PPR.
- Precision yield and wetland mapp data is used to estimate the effects of wetlands on crop yields and farm finances.
- Wetlands and their buffer zones have substantial negative effects on crop yields in the Prairie Pothole Region.
- Wetland drainage can mitigate these yield effects, providing further incentive for drainage in the PPR.
- The financial incentives for wetland drainage vary with respect to soil zone, technology use, and annual precipitation.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Wetland drainage has become an increasingly important conservation issue in the Prairie Pothole region of North America. Financial incentives for annual crop production have driven wetland drainage for decades, and the removal of wetlands has detrimentally impacted key wetland ecosystem services such as wildlife habitat and carbon sequestration. Past studies which model the farmer's decision to drain wetlands often operate on the assumption that drained wetlands will produce similar yields to upland regions of the field.
OBJECTIVE: Our objective is to estimate the effects of wetlands and their buffer zones on crop yields, farm financial performance, and incentives for wetland drainage in the Prairie Pothole Region.
METHODS: We combine precision yield data and detailed wetland mapping data from 36 fields in the Black and Dark Brown soil zones of Saskatchewan, Canada to estimate the agronomic impacts of wetlands and their buffer areas on crop yields. Then, we incorporate these yield effects into a farm accounting model with three wetland drainage scenarios to estimate the annual per cultivated acre net benefits of wetland drainage in the study area, and compare these results to those estimated without wetland yield effects.
RESULTS: We find that yields in wetland basins are relatively lower than the field's average yield, with substantial variability with respect to crop type, soil zone, and annual precipitation. Wetland drainage can mitigate

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these yield effects, but yields in drained wetland basins still fail to meet the field average yield. These yield effects can extend more than 50 m beyond the wetland boundary. We find that these effects substantially impact the net benefits of wetland drainage. The returns from wetland drainage increase when yield effects are considered. On average, full wetland drainage within the study area increases net benefits by $17 to $33 per cultivated acre relative to full wetland restoration.

SIGNIFICANCE: The results demonstrate the importance of considering wetland and buffer zone yield effects in wetland drainage decisions, improve our understanding of wetland costs, and potentially inform policy development and the design of incentives for wetland conservation in agricultural landscapes.

1. Introduction

Annual crop production has transformed the landscape of the Prairie Pothole Region (PPR) in North America over the last century. The PPR is characterized by the presence of depressions or pothole wetlands that dot the terrain, and includes Canada's Prairie Provinces – Alberta, Saskatchewan, and Manitoba – alongside several northern American states. These low-lying areas were formed by retreating glaciers at the end of the last ice age, and are hydrated with snowmelt each spring, serving a crucial role in the ecology of the region by providing habitat for numerous species, alongside other ecosystem services (Doherty et al., 2018). These wetlands that once defined the landscape of the PPR have been disappearing over time, often drained and converted to farmland, to increase agricultural productivity on a given parcel of land. Drainage activity is ongoing across the PPR. However, the literature reports a wide range in estimates of PPR wetland area lost to drainage, due to a lack of reliable data, varying definitions of loss and minimum wetland size, and timeframe constraints (Waz and Creed, 2017). Wetland drainage can have wide-ranging effects on the landscape, resulting in changes to watershed storage capacity, flood magnitude, nutrient distribution, groundwater recharge, salinity, biodiversity and habitat loss, and carbon sequestration (Baulch et al., 2021).

Although these wetlands provide immense value in ecosystem and recreational services (Pattison et al., 2011; Vickruck et al., 2019), the benefits that PPR wetlands afford to society are rarely reflected in representative markets and therefore not incorporated by landowners. The opportunity costs of forgone crop production associated with wetland preservation on private property motivate decisions to drain or farm through these areas (Gelso et al., 2008; Cortus et al., 2011; De Laporte, 2014; Clare et al., 2021; Asare et al., 2022). Further, wetlands can be a source of direct costs to landowners and producers. Nuisance and overlap costs arise as farmers spend increased time, fuel, and agricultural inputs navigating around wetlands when seeding, spraying, and harvesting (De Laporte, 2014). These effects are exacerbated when wetlands are dispersed throughout the field, severely reducing the maneuverability of machinery (Gelso et al., 2008). Wetland drainage can reduce nuisance and overlap costs, subsequently increasing efficiency at the field level (Clare et al., 2021). Finally, although wetlands can provide habitat for beneficial species (Vickruck et al., 2019; Robinson et al., 2022), they can also house pests and wildlife which damage crops (Taylor et al., 1993). Therefore, wetland drainage can increase both raw productivity by increasing the landowner's arable land, and production efficiency via nuisance and overlap cost reduction. Landowners in this region have also reported that undrained wetlands can impact crop productivity (Clare et al., 2021); however, these effects have rarely been considered or quantified in past economic research.

While a growing body of work on the topic has improved our understanding of the economic and ecological implications of wetland drainage, much of this research relies on the assumption that wetland areas and surrounding buffer zones will yield at levels similar to the remainder of the field. Realistically, yields in and around wetlands will be at least partially dependent upon the agronomic conditions imposed by the wetland, whereas yields in upland areas would not. This assumption creates a persistent gap in our understanding of how wetlands affect agricultural productivity and incentives for drainage.

The purpose of this paper is to use precision agricultural yield and wetland mapping data in Saskatchewan, Canada to estimate the effects of wetland drainage on annual crop productivity inside wetland areas and their surrounding buffer zones. We then include these effects in a farm financial analysis to assess the impacts of wetland drainage on farm profitability. This research will foster a stronger understanding of the relationships between wetlands, crop yields, and farm financial performance, improving on-farm wetland management and conservation efforts in the Prairie Pothole Region.

The paper is organized as follows. First, we present a brief literature review, highlighting previous studies which investigate the incentives for wetland drainage in the PPR. Next, annual crop yield and wetland characteristic data is explained. Descriptive analyses are then performed to visualize how wetlands affected crop yields throughout the study period. This descriptive analysis complements and informs the statistical analysis, where a series of OLS regression models quantify the effects of wetland drainage on yields in and around wetland basins. These estimates are used in a farm financial analysis which estimates the per cultivated acre net profits of wetland drainage under three drainage scenarios, which reveal the costs of wetland conservation. These estimates are compared with those under a "no yield response" assumption, where wetland basins and their buffer zones are assumed to yield the same as the remaining field. The differences in net benefit estimates between yield assumptions demonstrate the importance of accounting for wetland yield effects when assessing the costs and benefits of wetland drainage. Finally, results are discussed, research limitations are noted, and directions for future research are identified.

2. Incentives for wetland drainage in the north American prairies

The relationship between wetlands and farm financial returns has been studied extensively in the last few decades, typically to quantify the costs of conserving wetlands or better understand the driving forces of wetland drainage and conversion to cropland. Cortus et al. (2011) employed a cash flow simulation model based on a representative farm in Eastern Saskatchewan, finding net economic benefits of wetland drainage to range from $28 to $41 per hectare of drained wetland. Meanwhile, Asare et al. (2022) found average annual net benefits of wetland drainage to be $187 per hectare, though the authors posit that this value is on the high end of the spectrum.

Despite clear financial incentives on aggregate, there is significant spatial and temporal heterogeneity in the economic costs imposed by wetlands (Yang et al., 2016). Rashford et al. (2011) found that rates of wetland conversion are significantly higher in areas with higher soil quality. A considerable portion of wetlands are currently economically infeasible to drain, but high commodity prices can still result in very high rates of drainage (De Laporte, 2014; Asare et al., 2022). Farmer participation in Business Risk Management Programs such as margin and crop insurance further promote wetland drainage and are particularly influential when commodity prices are low (Cortus et al., 2009; Jeffrey et al., 2017). Therefore, when commodity prices are strong, wetland drainage is likely to be prominent, even in areas with relatively poor agronomic conditions.

The effects of wetlands on farming activities are also reflected in land

values. Using a stated preference survey of farmers in Kansas, Gelso et al. (2008) found that the value of cropland declines when the number of wetland basins inside the field increase, and as wetland permanence increases. Wetland consolidation and drainage can reduce costs and increase returns, and these improvements in profitability are capitalized into land values. Using a hedonic price approach, Schultz and Taff (2004) also found wetland acreage to reduce land values in North Dakota. Further, the negative effects of wetlands on land prices have increased over time, implying that returns to wetland conversion have increased concurrently (Lawley, 2014).

To our knowledge, only one study directly addresses the effects of wetlands on agricultural productivity in the Prairie Pothole Region at the sub-field level. Clare et al. (2021) investigated the effects of wetlands on canola production in Alberta during the 2019 growing season using real farm input and yield data for a comprehensive financial analysis. The authors found a high degree of variability in profitability associated with the presence of wetlands, with many individual basins yielding at a financial loss. The present research complements Clare et al. (2021) by incorporating a larger sample of fields over several years, with four different crop types. Although patterns of wetland loss are indicative of clear financial incentives for wetland drainage, research which directly analyses the agricultural productivity of wetland areas in the PPR remains limited.

3. Data

3.1. Study area

The study area is in the Prairie Pothole region of Saskatchewan, Canada. This is an agriculturally productive region with Chernozemic soils. Our study includes fields from both the Black and Dark Brown soil zones. The agronomic differences between these two soil zones enables an analysis of spatial heterogeneity in the investigation of wetland yield effects in the Prairie Pothole region. The Black soil zone encompasses a wide swathe of land in central and eastern Saskatchewan, while the Dark Brown soil zone lies directly to the south and west of the Black soil zone (Bedard-Haughn et al., 2018). There tends to be climatic and soil characteristic differences between the Dark Brown and Black soil zones. For example, the Black soil zone is slightly cooler, with more precipitation than the Dark Brown soil zone (Pennock et al., 2011).

3.2. Precision yield data

We use precision yield data to quantify the effects of wetland areas and buffer zones on field crop productivity. Advances in agricultural technology in recent decades have resulted in the development of precision agriculture production methods. Precision agriculture technology allows producers to spatially track input use and yields throughout the field to optimize technical and economic efficiency at the sub-field level. Precision agriculture techniques have become increasingly popular among Canadian farmers – 84 % of surveyed farmers were utilizing some sort of this technology by 2017, and 60 % were using GPS yield monitors (Steel, 2017). Moreover, Saskatchewan farms have the highest adoption rates for many precision agriculture technologies, including auto-steer, variable-rate input application, and GIS field mapping (St. Pierre and Mhlanga, 2022). Increasingly widespread availability of precision agriculture data facilitates accurate and detailed farm financial analysis.

Precision yield data was captured by the combine in the sampled fields at the time of harvest. Yield monitors measured the weight and percent moisture of the crop to calculate a yield per unit area every three meters along each harvested path, which were typically 15 m apart. The combine yield data is then interpolated to produce a raster dataset of yield for the field, displaying how yield varies in different parts of the field. Each field is then divided into polygons consisting of upland acres, wetland basin acres, and wetland buffer zone acres, which facilitate analysis of the factors which drive yield variance among different areas within a field.

The data used in this analysis was collected from 36 fields in the Black and Dark Brown soil zones. The Black soil zone data represents 16 fields, containing 4689 acres over 7 years (2014–2020). Meanwhile, the data from the Dark Brown soil zone represents 20 fields, containing 10,433 acres, and spanning a timeframe of 4 years (2016–2019). The data was unbalanced, since not all fields had yield data for all of the study years. Although yield data is collected from a variety of crops, only four – wheat, canola, malt barley, and yellow peas – were considered, due to a lack of sufficient data for other crop types. Malt barley is considered in both soil zones in the descriptive analysis but is only considered in the Black soil zone in the statistical and financial analyses due to data limitations.

Several data cleaning and transformation steps were taken with the raw combine yield data. Many factors can impact the raw data across fields and seasons, including weather, inherent soil quality, crop type, field fertility history, annual fertility and weed control methods, and uncertainty about the calibration of yield monitors. To address these sources of variation, we relativize each polygon's yield data to be expressed as a percentage of the average yield in the field for that year. The field average yield is defined as the average yield of all seeded acres, including uplands and seeded wetlands, in a specific field. Therefore, a yield value of 80 in a given polygon implies that the yield in that area is 80 % of the field average in that year. The expression of yield in this way elicits direct comparisons of yields in wetlands and their buffer areas across fields, accounting for variation in fields, years, and agronomy. The drawback is that overall field averages will depend on the wetland and drainage extent in each field. Ultimately, these data cleaning and transformation procedures were deemed necessary to ensure spatial and temporal comparisons of yield across fields remained sound. We also remove some outliers from the data where yield exceeded 200 % of the field average (528 (0.66 %) in total), as these values are agronomically unlikely, and a likely result of machinery errors. There are 298 (0.37 %) instances of yield being equal to zero, which remain in the dataset. Steps were taken to ensure that unseeded areas were not included in the final dataset, though some may have been missed. Therefore, we assume that these areas simply did not yield any noticeable amount at the time of harvest.

3.3. Wetland and buffer zone data

Wetland inventory of the study area was collected using Canadian Wetland Inventory (CWI) methods (Boychuk et al., 2014). GIS polygons were used to delineate each wetland area. In total, 6179 wetland polygons were identified on the 36 fields over the study timeframe, enabling multiple year-observations for each wetland, though some fields only have one year of data. Further, the number of wetland observations varies year-by-year and field-by-field due to crop yield data availability and removal of crop types for which there was insufficient data. While the area of each wetland basin is static through time, there is some variation in data availability within and for each field over time.

Yield polygons from a 50 m buffer zone surrounding the each wetland basin are also identified. Yield polygons in the buffer zone are separated into 5 m increments beginning at the edge of the wetland basin and extending to 50 m away. The dataset contains 73,425 buffer zone polygons across all fields. Yield data from outside the wetland and wetland buffer zone areas is not included in the analysis, but since yield is specified as the percentage of field average, the yield variable within the wetland and buffer polygons serves as a comparison to average of the entire field. In total, the data from the Black soil zone contains 49,453 wetland and buffer zone observations, while the Dark Brown soil zone data contains 30,151 wetland and buffer zone observations. Fig. 1 displays how wetland buffer zone polygons are accounted for and explains how cultivated and total field area are calculated.

The wetlands in the dataset are categorized and classified based on a wetland impact code (Boychuk et al., 2014). Wetland impact code

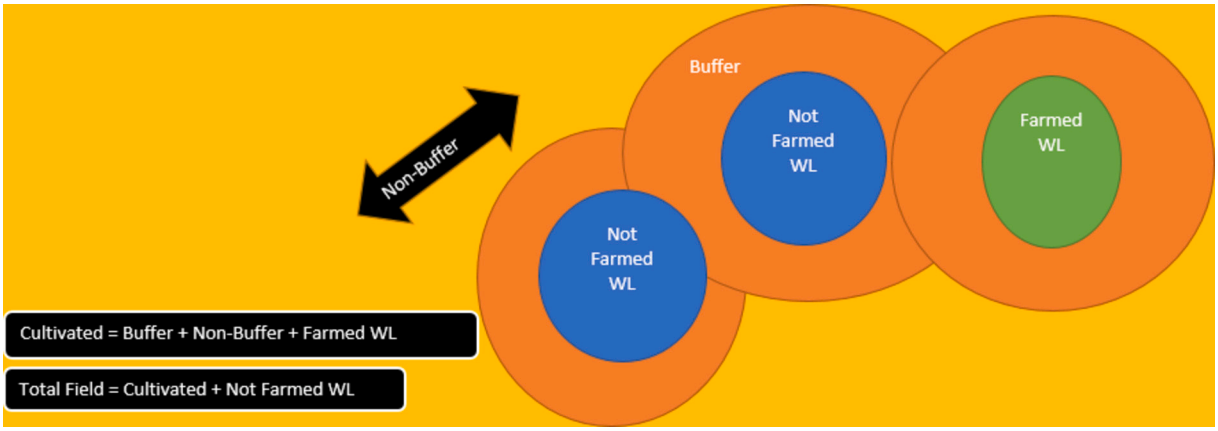


Fig. 1. Wetland buffer zone overlap and field definitions.

describes the level of drainage a given wetland has been subjected to. We use four of the levels most relevant to our study area[1] and present their definitions in Table 1. Impact codes were only categorized once, and thus are assumed to remain static over the period of the study.

4. Empirical yield analysis

4.1. Descriptive analysis

We present a descriptive analysis of the data prior to the empirical and financial modelling, pooled across both the Black and Dark Brown soil zones. Fig. 2 illustrates the distribution of relative crop yields in drained and farmed wetland basins. On average, yields in these sub-field regions are lower than the field average. However, there is a noticeable clustering of yields between 100 and 125 % of the field average, indicating that under certain conditions, wetland basins can be quite productive. These yield visualizations seem to indicate that wetland basins, even when drained, are associated with a high degree of variability in yield. While we are not able to compare these yield distributions to those of upland areas of the field, the relatively flat distributions inside wetland basins indicate a wide range of outcomes over the study timeframe.

Fig. 3 reports the distribution of relative crop yields in the 50-m buffer zone polygons. The yield distribution in the buffer zone differs from that within the wetland basin, approaching a more normal distribution. Buffer zones do not cause the same degree of yield variation as wetland basins. Mean yields adjacent to intact and partially drained wetlands are less than those inside farmed and drained wetland basins, indicative of the spatial extent of negative yield responses associated with undrained wetlands. These effects are less pronounced in buffer zones surrounding farmed and drained wetland basins, where mean yields nearly meet the field average.

Farmers should generally expect reduced yields and increased yield volatility in areas close to wetland basins, with these effects dissipating through the buffer zone into the rest of the field (see Appendix B for summary statistics of yields for wetland basins and each 5 m buffer zone increment). However, this convergence with upland field characteristics may not be perfectly linear. Field edge effects can be highly variable, and differ among crop types (Robinson et al., 2022). Figs. 4 through 7 illustrate how average yields change through the 50 m buffer zone, organized by impact code and crop, and depending on annual moisture.

Figs. 4 and 5 display the buffer zone yield pattern with respect to impact code. Yields where 'Distance from Wetland (m)' = 0 correspond to yields within wetland basins; yields where 'Distance from Wetland (m)' = 5 correspond to the first 5 m increment of the wetland buffer zone, and so on until 50 m beyond the wetland basin boundary. On average, yields surrounding farmed and drained wetlands reach 100 % of the field average at the outside margin of the 50 m buffer zone. As expected, the average yields extending from partially drained and intact wetlands are relatively low, only reaching around 85 % of the field average at 50 m outside the wetland basin. Trends differ based on annual moisture.[2] In relatively dry years, yields across impact codes are noticeably higher directly adjacent to the wetland basin, before dropping off and rising again around 20 m from the wetland basin. This trend may be due to greater soil moisture availability within or next to wetland basins, which can boost crop yields under dry conditions. Moreover, farmed and drained buffer zone yields reach or exceed the field average 50 m from the wetland in dry years. In wet years, average yields within and next to the wetland basin are very low, but steadily increase through the buffer zone. This could be due to issues of excess soil moisture near the wetland basin, the effects of which rapidly taper off through the buffer zone. Yields alongside farmed and drained wetlands increase rapidly, nearly reaching the field average by the perimeter of the 50 m buffer zone in wet years.

Figs. 6 and 7 further granularize wetland yield effects by displaying the average yield patterns through the buffer zone by crop and impact code. We observe a similar pattern, where crop yields in and around drained and farmed wetlands noticeably exceed those in and around undrained wetlands across crop types. Yellow peas perform particularly poorly in and around wetland areas, relative to the remaining field. This

Table 1
Definitions of wetland impact code variable.

Impact Code	Definition
Intact	No evidence of drainage.
Partially drained	The water level has been lowered, but the soil contains enough moisture to support hydrophytes (water plants).
Farmed	The soil area has been altered for crop production but remains undrained. If farming is discontinued, hydrophytes will return.
Completely drained	The soil surface has been altered for crop production, and the water level has been lowered enough to generally support annual crop production.

[1] The two impact codes we do not consider are not represented in the data and include constructed wetlands (the soil area has been excavated to create a water-holding basin) and partially filled wetlands (the basin shows evidence of clearing).

[2] Precipitation data was obtained from the Government of Canada's (2022) historical weather and climate data and McKenzie (2019), and is described in Appendix C.

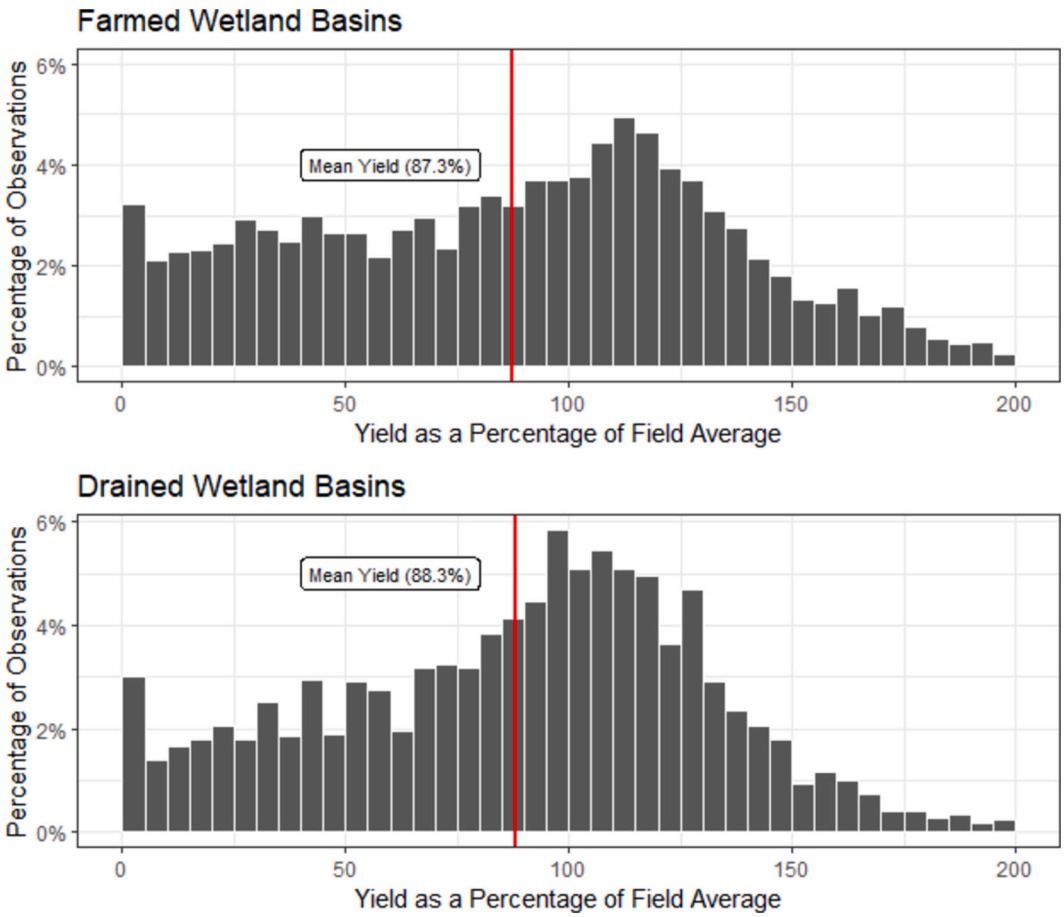


Fig. 2. Yield distribution within wetland basins.
Notes: While there is yield data for polygons inside partially drained and intact wetlands, they are assumed to be unseeded for this analysis. Of the 6179 wetland observations in the data, 3231 are farmed, 1793 are drained, and 1155 are unseeded.

result aligns with agronomic expectations, as peas tend to be sensitive to saturated and saline soils, characteristics often associated with wetlands (Agriculture and Agri-Food Canada, 2008). We again observe differing trends with respect to precipitation. Crop yields are generally higher in and around wetland basins in dry years across crops, with canola in particular yielding above the field average throughout most of the buffer zone adjacent to drained and farmed wetland basins. While yields are lower in relatively wet years, trends indicate substantial improvement through the buffer zone, particularly 30 m from the wetland basin and beyond.

In summary, wetland basins tend to result in reduced yields relative to the rest of the field, and these negative yield effects extend into surrounding buffer zones. Wetland basins and their buffer zones are also associated with a substantial amount of yield variability, driven in part by annual precipitation and crop choice. However, the available data limits our ability to compare this variance with upland areas. Wetland drainage increases yields within the wetland areas and mitigates the extent of these effects and variability therein further into the field. While average yields are still reduced within drained wetlands relative to the rest of the field, the mitigation of wetland-associated yield reductions beyond the wetland itself motivates further analysis to quantify these effects of wetland drainage on farm financial performance. The stark difference in buffer zone yield trends with respect to wetland drainage activities is one yet to be captured from an economic perspective and is emblematic of a clear financial incentive for wetland drainage from the farmer's perspective.

4.2. Statistical analysis

To estimate the impacts of wetlands and their buffers on crop yields in the Black and Dark Brown soil zones, we specify a linear regression model where yield is a function of a full factorial interaction of crop type, wetland impact code, and wetland/buffer location, which is estimated separately by soil zone. These estimates expand upon the descriptive analysis and serve as inputs for quantifying the effects of wetland drainage on farm financial performance. The estimated model parameters can be interpreted as the average yield responses for each crop within or surrounding wetland basins subject to varying levels of drainage activity. We do not incorporate field or year fixed effects and use a pooled Ordinary Least Squares estimator to recover the model parameters because the dependent variable is yield as a percentage of the field average for the year, so unobserved, time-varying variables at the field level are already controlled for in the analysis.

Let yield as a percentage of field average in polygon i, in field j, during year t be modelled as:

$$\frac{Yield_{ijt}}{Field\ Average\ Yield_{jt}} = \beta c_{jt} x_i z_i + \varepsilon_{ijt} \tag{4.1}$$

where c_{jt} is the crop planted in field j in year t, x_i is the impact code of polygon i, and z_i is the location of polygon i – either inside a wetland basin or in a surrounding buffer zone, and ε_{ijt} is an error term capturing remaining unobserved factors. The estimated parameter vector β is interpreted as the average yield response of each crop, in each impact code, in either a wetland basin or buffer zone as a percentage of the field average. Standard errors are clustered at the year and field level to

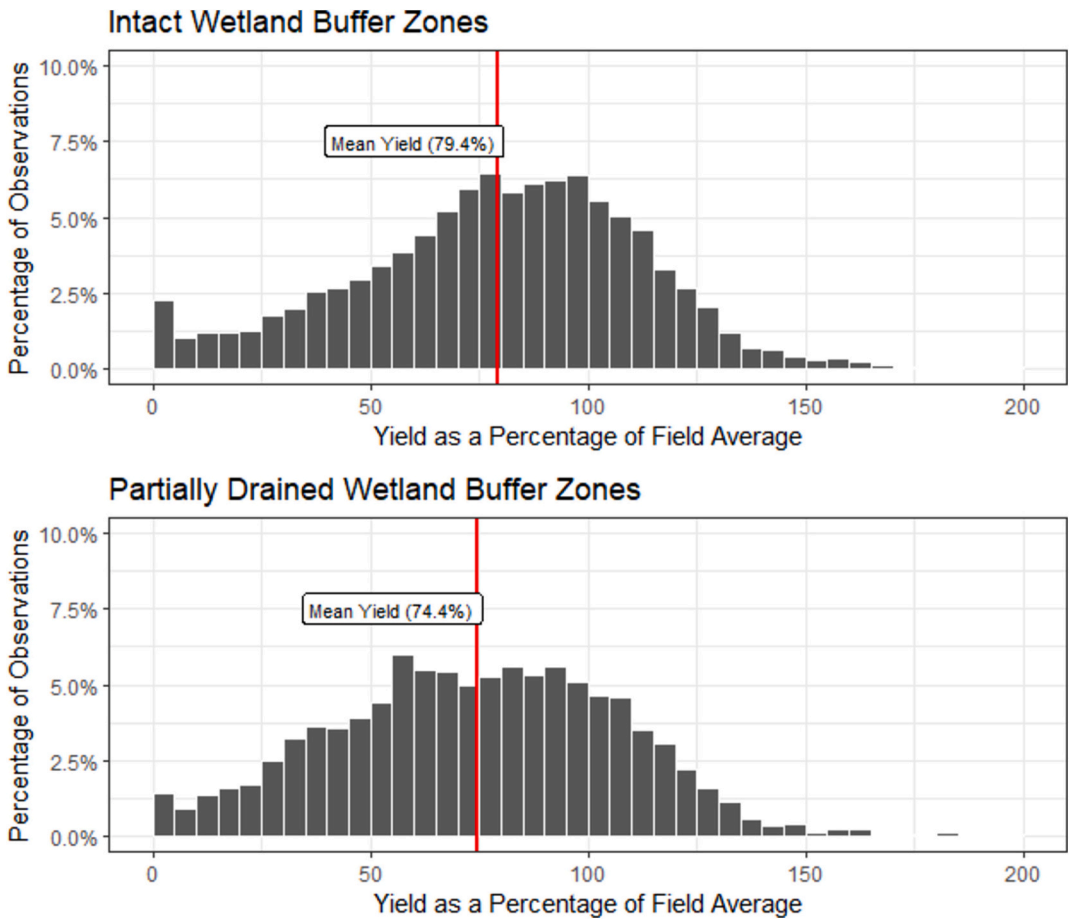


Fig. 3. Yield distribution within 50 m wetland buffer zones.
Notes: Of the 73,425 buffer zone observations in the data, 14,916 (20 %) are adjacent to intact wetlands, 3624 (5 %) are adjacent to partially drained wetlands, 35,953 (49 %) are adjacent to farmed wetlands, and 18,932 (26 %) are adjacent to drained wetlands.

address the fact that multiple polygon data points are from the same field in the same year. Summary statistics of the model parameters are presented in Table 2.

Fig. 8 displays the average yield response by impact code and crop for the Black and Dark Brown soil zones. Error bars correspond to the standard error of each average yield response, exhibiting the variability in crop yields associated with different drainage conditions in wetland basins and their buffer zones. Since intact and partially drained wetland basins are assumed to be unseeded, average yield response in these areas is zero with no variance in yield and not displayed.

The model uses observational data, and therefore caution is warranted in assigning causality to the estimated yield effects. The yield response depends on the wetland impact code but also other farmer decisions that may be correlated with wetland extent. For example, farmers may use a different level of inputs on fields with many wetlands but we assume similar per acre costs and assume that the yield response for further wetland drainage or restoration is equal to the estimated difference in yields between impact codes in the data, which is based on past decisions. While factors such as crop input use (fertilizers, pesticides, etc.), soil quality, climatic factors such as precipitation and temperature, and pest pressure, among others, are key determinants of yield, data on these factors is unavailable. However, given that crop yield is expressed as a percentage of the field's average, the potential confounding nature of these factors is addressed to the extent that these factors have similar effects across the field. The use of variable rate input application can result in a high degree of subfield input use variability, so it is assumed that input application rates maintain a proportional relationship with crop yields across polygons and fields. The degree to which variable rate application technology was utilized across the fields and years in the dataset is unknown.

The model results align with and quantify the outcomes of the descriptive analysis. Yield response is consistently closer to the field average in and around farmed and drained wetlands than partially drained or intact wetlands. In the Dark Brown soil zone, yields in these areas often exceed the field average. Standard error estimates tend to be higher inside wetland basins than in the buffer zones, illustrating the wide range of agronomic conditions wetlands can impose on crop production. Predictably, yellow peas incur a significant reduction in yield in and around wetland basins relative to other crops. While the economic incentives of wetland drainage – particularly those pertaining to increased cropped acreage and cost reductions – have been rigorously investigated, these yield effects in farmed and drained wetlands, relative to those left intact, are illustrative of further incentives for farm wetland drainage.

5. Farm financial analysis

5.1. Description of approach

We integrate the yield response results from the empirical yield analysis into a farm financial analysis to reveal how farm financial performance changes when accounting for the yield effects of wetlands and their buffer zones. We focus on changes in per cultivated acre farm profitability and consider three wetland drainage scenarios: 1) a current

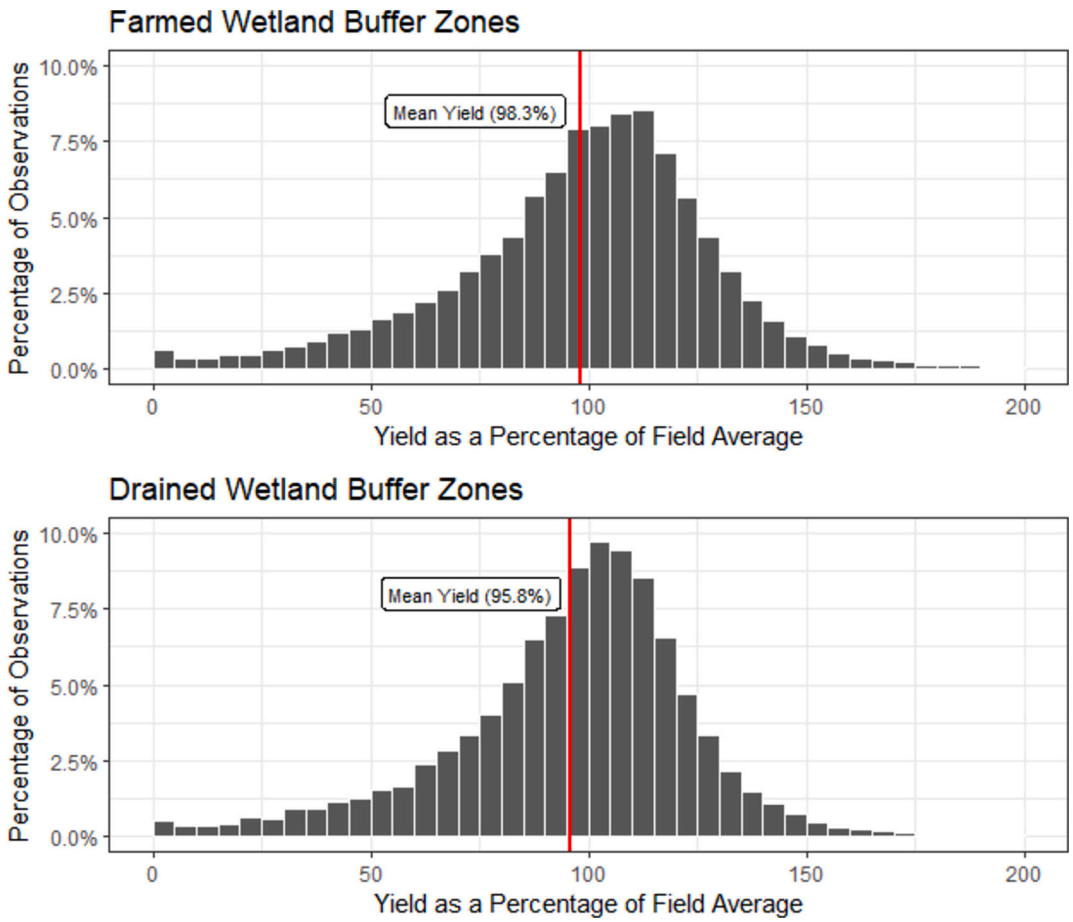


Fig. 3. (*continued*).

conditions scenario, 2) a full drainage scenario, where all wetland basins feasible to drain are assumed to be drained (30 % of wetland area in the sample is Impact Code 2 wetlands, which are farmed but not drained), and 3) a restoration scenario, where all wetland basins (besides Impact Code 2) are assumed to be intact. The differences in net benefit estimates among these three scenarios reveal the private costs of wetland conservation, or the financial incentives for wetland drainage. The financial analysis is conducted under two crop yield assumptions: one where yield responses to wetland and buffer zone conditions are accounted for, and one where no yield response is considered, meaning wetland and buffer zone yields are assumed equal to the field average yield. This elicits a comparison of financial outcomes between the commonly held yield assumptions in previous research, and the yield responses observed over the span of the study, highlighting a disparity with implications for future research. We expand the analysis to compare the differences in profitability across drainage scenarios in relatively wet and dry years, and investigate how estimates change with the adoption of sectional control technology on field operations equipment.

The revenues and costs of farm production for the financial analysis are derived from the 2022 Saskatchewan Crop Planning Guide (Government of Saskatchewan, 2022). Crop input cost values are based on those required to achieve 80th percentile target yields, unique to the agronomic conditions of the Dark Brown and Black soil zones respectively. These target yields are also assumed to be the field average yields. While crop yields may not always achieve the 80th percentile target in practice, this target yield was selected to represent the realistic per acre input costs a Saskatchewan farmer incurs. The analysis considers revenues from crop production under both yield response assumptions, the annual costs of crop production, input overlap costs (as a percentage of crop input costs), nuisance costs (as a percentage of machinery operating costs), and wetland drainage costs. The financial analysis is calculated separately for the Black and Dark Brown soil zones and specified as follows:

Per Cultivated Acre Farm Net Benefits

$$= \frac{\sum (P_c Y_{cs} \beta_{cxzs}) - \sum (C_{cs} + O_{cs} + N_s + D_s)}{Cultivated\ Acres_s}$$

Where P_c is the farm gate price for crop c, Y_{cs} is the 80th percentile target yield for crop c under drainage scenario s, β_{cxzs} is the yield response for crop c, impact code x, and location z under drainage scenario s (equal to 1 for upland yields and wetlands/buffer zones in the no yield response assumption), C_{cs} is the annual input costs associated with achieving the target yield under drainage scenario s, O_{cs} is the overlap costs of crop c under drainage scenario s, N_s is the nuisance costs under drainage scenario s, and D_s is the drainage costs associated with drainage scenario s.

The results of this analysis are expressed as the annual per cultivated acre net benefits of farming[3] under each wetland drainage scenario. This makes net benefit estimates comparable across fields and drainage scenarios, independent from arable land constraints that may vary by field and drainage scenario. Further, the expression of results in per cultivated acre terms normalizes the differences in cultivated acres among drainage scenarios and removes the opportunity cost savings inherent to wetland drainage from the equation, allowing for a focus on the direct costs of wetland conservation among the three drainage scenarios.

[3] Not all fields are from the same farming operation, but the results can be interpreted as the net benefits of a farm operation containing the sampled fields.

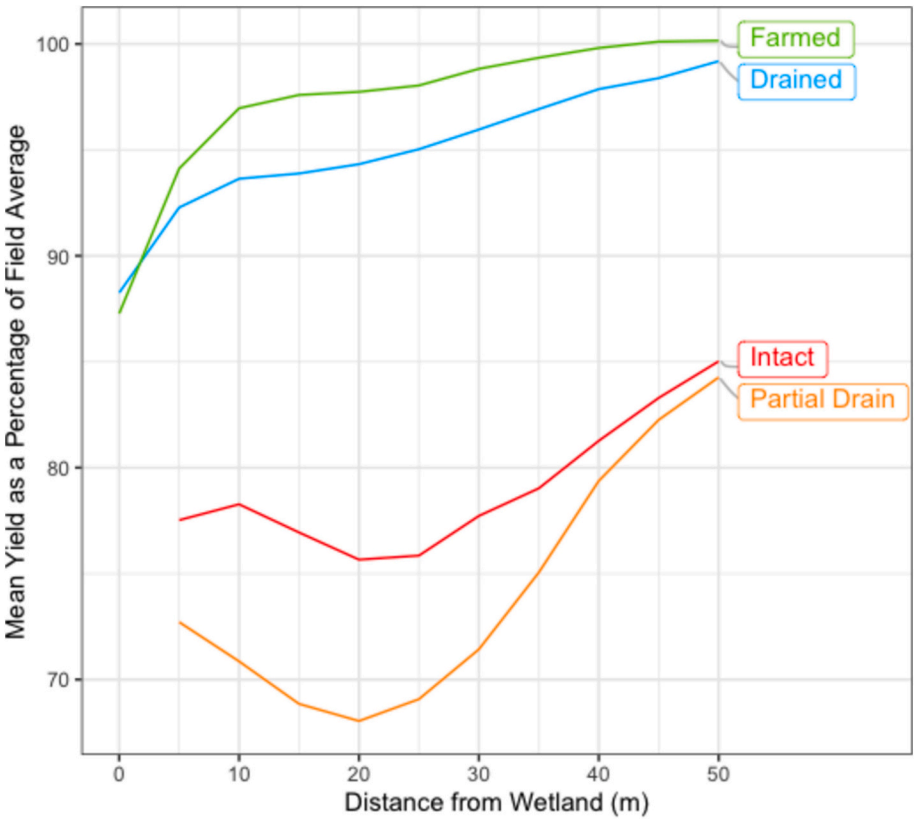


Fig. 4. Average yields by wetland impact code within wetland basins and through the 50 m buffer zone, $n = 78{,}449$.

Notably, much of the previous literature expresses farm net benefit estimates in dollars per wetland acre, rather than per cultivated acre, meaning that monetary values may not be directly comparable to other research. The analysis is conducted on an annual basis and thus we ignore any yield and financial dynamics from drainage decisions. While up to seven field-years are included in the yield effect calculations, our results should be interpreted as the farm financial outcomes in a single year, based on the yield response estimates obtained through the study time frame and the crop yield, price and input cost assumptions outlined in the 2022 Saskatchewan Crop Planning Guide.

The analysis assumes an equal crop rotation; therefore, wheat, canola, yellow peas, and malt barley each make up 25 % of Black soil zone acres, while acres in the Dark Brown soil zone are equal parts wheat, canola, and yellow peas. The costs and revenues associated with field crop production are detailed in Appendix E. Associated agronomic assumptions are presented in Appendix F.

Drainage, overlap, and nuisance costs are included in the financial analysis to comprehensively evaluate the costs and benefits of wetland drainage. Drainage costs are derived from a digital elevation model (DEM) of 148 drainage ditches identified in the fields in the sample. To estimate annual wetland drainage costs, a relationship between ditch excavation volume and area of wetland drained is established and factored together with drainage costs of $\$4/m^3$ of ditch material excavated and a borrowing rate of 5 % amortized over 25 years, resulting in an annual cost of $267/acre drained. These costs are at the high end of the spectrum for our study area.

Overlap and nuisance costs are developed using similar processes. Field path data, implement width, and total seeded/sprayed area are used to determine the percentage of input overlap on each field. Then, a linear relationship between overlap percentage and the percentage of the field represented as intact and partially drained wetland is estimated and used as an index to determine the overlap cost percentage. This value is applied to crop input costs and machinery expenses to estimate input overlap and nuisance costs respectively.

Scenarios involving sectional control on field operations equipment are also considered. Sectional control reduces input overlap by turning off individual portions of farm implements when farmers navigate around obstacles and field boundaries. Sectional control technology is becoming increasingly available to Canadian farmers; 73 % of surveyed farmers used this technology on at least one piece of equipment in 2017 (Steel, 2017). This analysis assumes that when sectional control is used, overlap with previous input applications is decreased such that input overlap costs are 8 times less than they would be without sectional control. This estimate is based on the suggestion of the Water Security Agency for the average effectiveness of sectional control technology in Saskatchewan. Practically, the effectiveness of sectional control in reducing overlap costs will vary based on a number of factors, including implement size, row spacing, field size, and the size and shape of wetlands and other obstacles (Gaetz and Lung, 2020; Barker, 2022). The relative benefits of wetland drainage between drainage scenarios are expected to be reduced when sectional control technology is adopted.[4]

There are several instances where buffer zone polygons are within 100 m of one another, therefore overlapping. To alleviate this issue, a buffer correction index was developed by dividing the buffer area (total field area, less total wetland area and non-buffer area) by the area of all buffer zone polygons. This index was used to adjust the buffer areas associated with each wetland impact code. Wetland and corrected buffer zone areas are displayed in Appendix G.

The yield response assumptions are applied to three different

[4] While machinery costs are included according to Saskatchewan Crop Planning Guide (2022) estimates, we do not alter these cost estimates among the sectional control and no sectional control scenarios. Incremental costs of sectional control are a small portion of farm machinery costs. Given the prevalence of sectional control use in our study area, this likely implies that the net benefits without sectional control use are slightly underestimated.

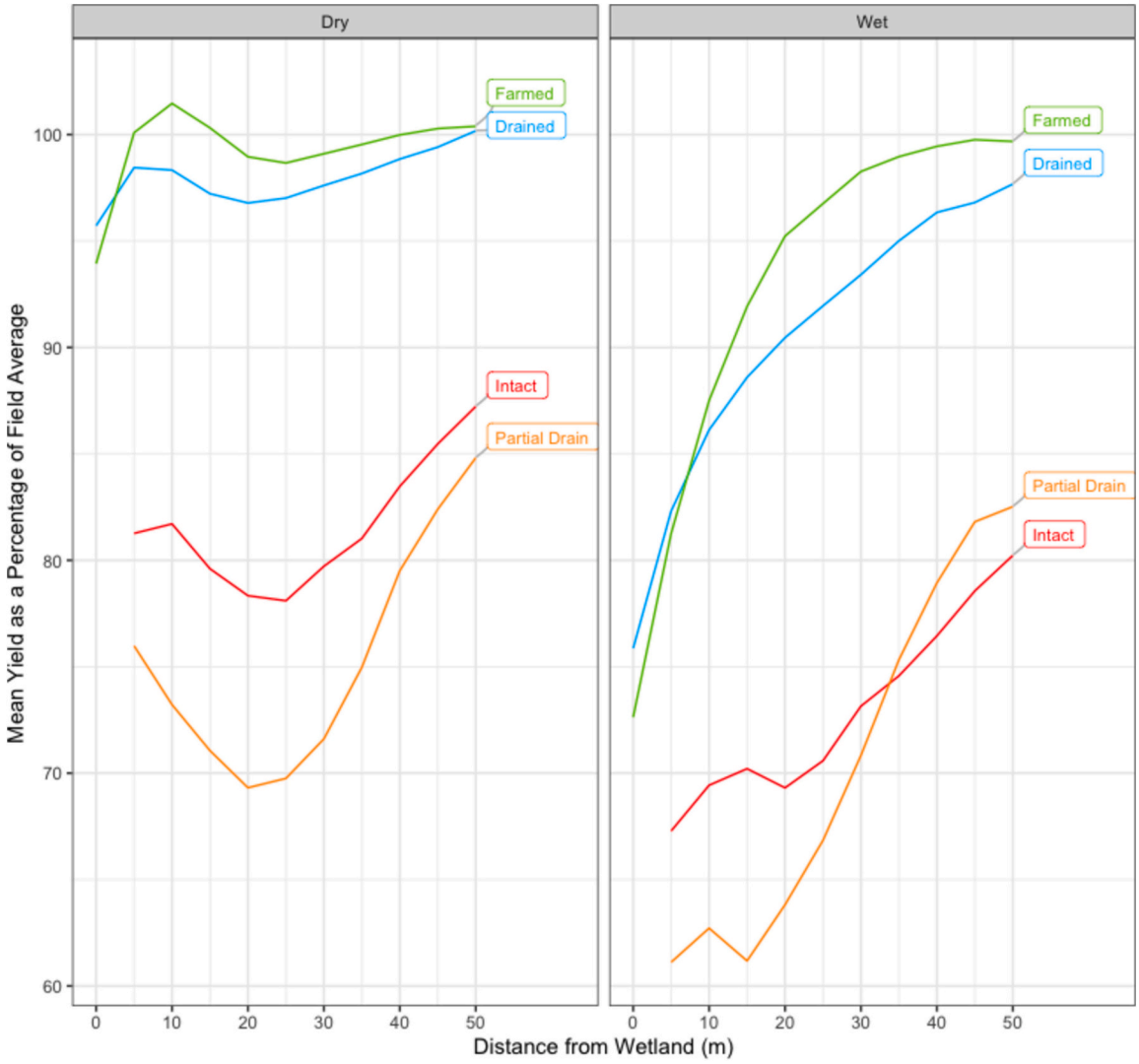


Fig. 5. Average yields by wetland impact code within wetland basins and through the 50 m buffer zone, during relatively wet and dry years, $n = 78{,}449$.

wetland drainage scenarios. Descriptions of each drainage scenario, and the yield response assumptions required for each scenario are found in Table 3. Intact and partially drained wetlands are always assumed to be unseeded, and yield responses associated with farmed wetlands remain unchanged across scenarios.

These drainage scenarios allow for the annual net benefits and financial incentives for wetland drainage to be assessed and compared across a broad spectrum of outcomes. The yield response matrix for each drainage scenario is displayed in Appendix H.

5.2. Farm financial analysis main results

Results from the farm financial analysis, under each yield assumption and drainage scenario, are displayed in Table 4. This set of results assumes the use of sectional control on seeding and spraying equipment, as this technology is commonly used across Saskatchewan farming operations. In total, the Black soil zone study area consists of 4689 acres, while the Dark Brown soil zone study area contains 10,433 acres. However, the amount of cultivated acres varies by scenario. For example, since all wetlands are either drained or farmed through in the full drainage scenario, 100 % of the acres in the study area are cultivated. Meanwhile, cultivated acres are reduced in the other two scenarios, since intact and partially drained wetlands are assumed to be unseeded.

Wetland yield response assumptions are responsible for substantial differences in farm net benefit estimates. When yield response effects are not considered, farm net benefit estimates are higher than those calculated under the yield response assumption, suggesting that previous estimations of the benefits of wetland drainage that do not consider yield effect costs could be similarly exaggerated. Further, full wetland drainage under this assumption results in decreased farm net benefits, an unexpected and counterintuitive result which implies that farmers are disincentivized to drain wetlands and does not align with observed trends of wetland drainage (Lloyd-Smith et al., 2020). This occurs because the costs of wetland drainage exceed nuisance and input overlap cost savings, making whole-farm wetland drainage financially infeasible.

When yield effects are accounted for, raw farm net benefit estimates are relatively lower, as yields tend to be below the field average (yield response = 1) in and around wetland basins. However, wetland drainage and restoration have more predictable effects on farm finances, where per cultivated acre net benefits increase under full drainage and decrease under full restoration. In other words, we obtain more predictable wetland conservation cost estimates. Thus, the farmer is incentivized to continue draining wetlands to increase farm-level profitability. Although drained wetlands tend to yield less than upland areas, drainage does improve yields relative to intact wetlands, therefore increasing crop yields and financial benefits across the operation. This

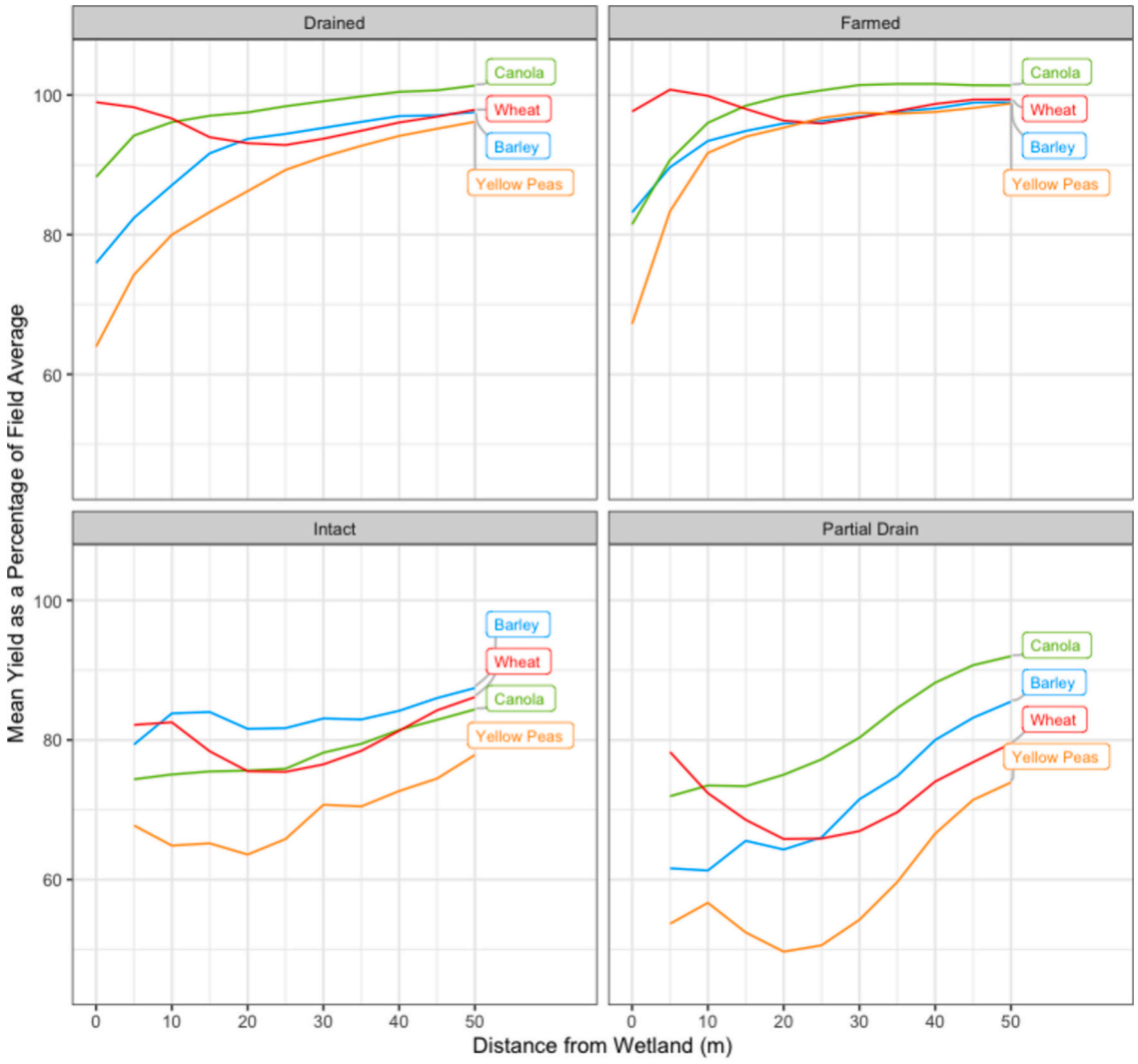


Fig. 6. Average yields by crop type and impact code from within wetland basins and through the 50 m buffer zone, n = 78,449.

relative yield improvement is not captured without the inclusion of yield responses, and the reductions in overlap and nuisance costs facilitated by drainage do not outweigh drainage costs without the consideration of yield effects. Therefore, yield effects are a necessary consideration to accurately model the farmer's incentives for wetland drainage.

Importantly, net benefit estimates are best interpreted in terms relative to one another, rather than as standalone values. Each estimate is dependent upon assumptions for baseline crop yields, commodity prices, and input use. Therefore, the relative magnitudes among estimates between the drainage scenarios, and yield assumptions are the most valuable results of this analysis.

5.3. *Impact of precipitation on wetland conservation economics*

The descriptive analysis reveals differing trends in crop yields within the wetland basin and across the 50 m buffer zone with respect to annual precipitation. To quantify these effects on farm profitability, we adjust the yield response matrices for relatively wet and dry years, where years with above average annual precipitation are denoted as a wet year, and years where annual precipitation is below average are considered dry. The regression from Section 4.2 was then run separately for both wet and dry years. Since the Dark Brown soil zone sample only consists of four years, net benefits are only presented for the Black soil zone to ensure an informative sample. Further, yellow peas are excluded from the analysis due to a lack of data in certain years. Since yellow peas tend to cost less to produce and yield relatively poorly in and around wetland basins, inclusion of yellow peas in the crop rotation would reduce per acre farm revenues and decrease per acre input costs across the crop rotation, resulting in an ambiguous effect on farm profits. Therefore, these results should only be interpreted as a comparison between relatively wet and dry conditions and are not directly comparable with the four-crop rotation results presented in Table 4. Precipitation data used in the analysis can be found in Appendix C, and results are presented in Table 5.

In the Black soil zone, wetland basins and their surrounding buffer zones are more profitable in relatively dry years. This result aligns with the yield effects observed in the descriptive analysis, where yields tend to be noticeably higher in the wetland basin and across the buffer zone in dry years. The relative lack of moisture in these years may mitigate some of the negative agronomic effects of wetland basins.

While changes in per acre profits across drainage scenarios are similar in both wet and dry years in absolute terms, the changes as a proportion of total net benefits is much higher in wet years, showing that financial incentives for wetland drainage are higher under relatively wet conditions. Wetland drainage can mitigate the adverse effects of wetlands on farm profitability in wet years, and these financial benefits are retained in drier years.

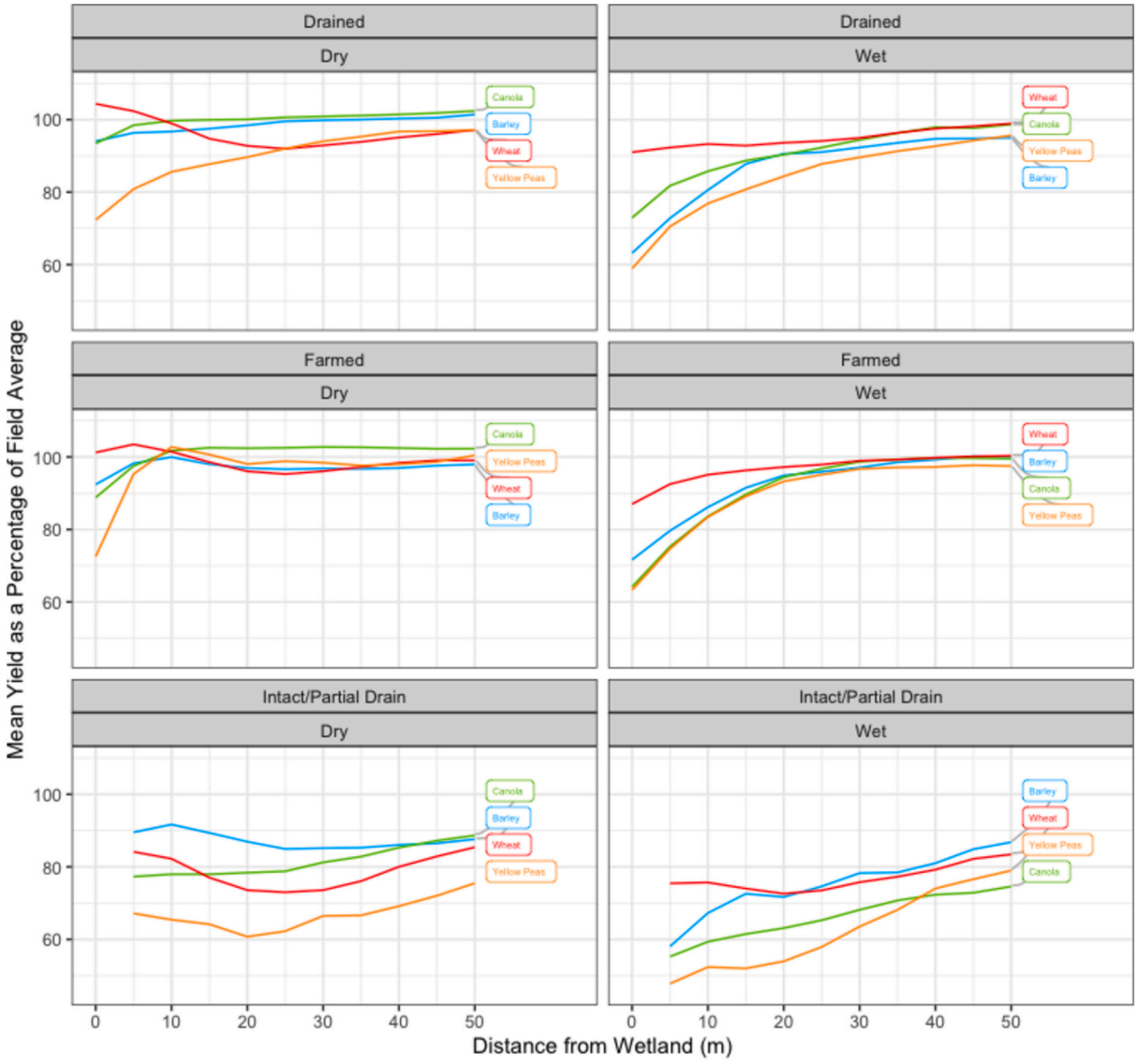


Fig. 7. Average yields by crop type and impact code from within wetland basins and through the 50 m buffer zone, during relatively wet and dry years, n = 78,449.

Table 2
Summary statistics for dependent and explanatory variables.

	Black Soil Zone	Dark Brown Soil Zone
Mean		
Number of Observations	49,453	27,669
Yield as a Percentage of Field Average[a]	90	93
Crop (proportion of observations)		
Barley	0.13	-[b]
Canola	0.55	0.29
Wheat	0.29	0.56
Yellow Peas	0.03	0.14
Wetland Impact Code (proportion of observations)		
Intact	0.21	0.17
Partially Drained	0.04	0.07
Farmed	0.49	0.50
Drained	0.26	0.26
Location (proportion of observations)		
Wetland Basin	0.08	0.08
Buffer Zone	0.92	0.92

[a] Yield in each polygon divided by the average yield in field. Percentage values are multiplied by 100, such that the range is 0 to 200.
[b] Malt barley is only considered for the Black soil zone due to data limitations.

5.4. Impact of technology on wetland conservation economics

To compare how farm financial analysis and the benefits of wetland drainage can be influenced by the adoption of farming technology, Table 6 displays the financial model results without the use of sectional control.

The ability to turn off input application when farm implements overlap reduces costs of production, thereby increasing farm net benefits. The results where sectional control is not utilized show broadly reduced profit margins. Financial incentives for wetland drainage are also altered when sectional control is forgone. For example, in the Black soil zone without sectional control, full wetland drainage increases profits by $29/cultivated acre relative to the current conditions. However, when sectional control is used, this disparity is only $14/cultivated acre. When yield effects are not considered, full restoration reduces net benefits as expected, as overlap cost savings from sectional control are no longer outweighed by wetland drainage costs. Overall, the adoption of sectional control technology reduces farmer incentives to drain wetlands as nuisance and overlap costs are mitigated. When all wetlands are drained or farmed through, sectional control does not affect net benefits of drainage, since farm machinery is not forced to navigate around wetland areas.

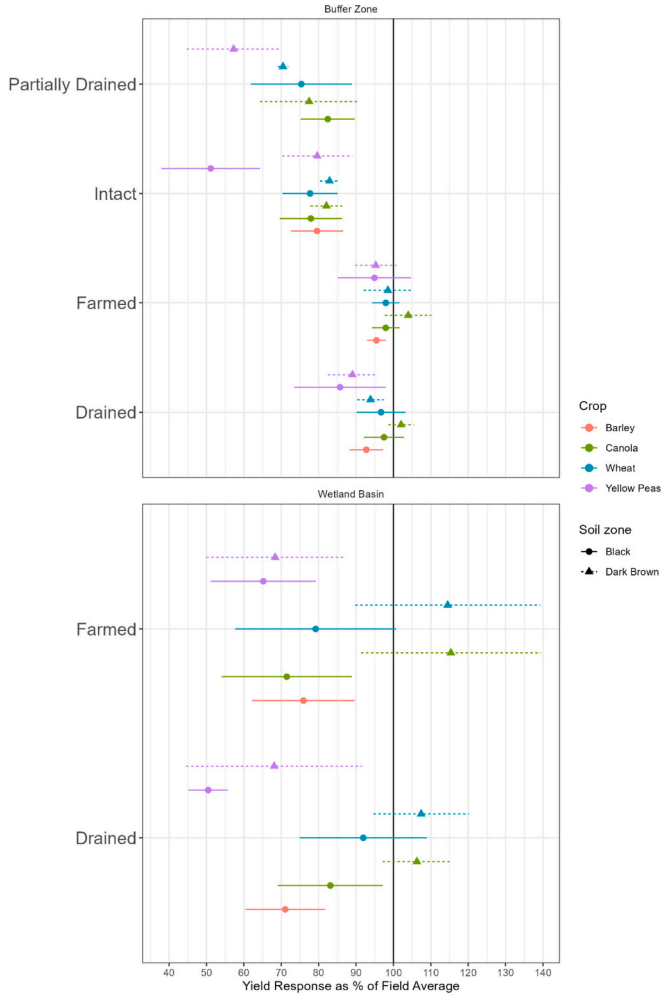


Fig. 8. Wetland and buffer area yield responses by crop, impact code, and soil zone.
Notes: The dots represent point estimates, and the error bars represent 95 % confidence intervals. Two parameters have standard errors of (near) zero due to limited observations and are not displayed. There are only 42 observations of barley (Dark Brown soil zone) in partially drained buffer zones and 57 observations of yellow peas (Black soil zone) in partially drained buffer zones. A table of estimates can be found in Appendix D.

6. Discussion and conclusions

The results of this study show that wetland yield effects have substantial impacts on farm productivity and profitability in the Prairie Pothole Region. While factors such as the opportunity cost of uncropped land and nuisance costs have long been known to drive wetland conversion to cropland (Cortus et al., 2011; De Laporte, 2014), the effects of wetlands and their buffer zones on crop yields have rarely been directly addressed in previous research (Clare et al., 2021). We include these effects into a farm financial analysis to illustrate the importance of these factors in incentivizing wetland drainage. The results of this study have wide-ranging implications for future economic analysis of agricultural wetlands in the Prairie Pothole Region.

Our results show that wetland and buffer zone yield effects reduce the net benefits of farming at the farm level. The differences in net benefit estimates between the two yield assumptions reveal the annual costs of wetland yield effects. For example, in the Black soil zone under the current conditions scenario, farm net benefits are $153 per cultivated acre without the yield effect, and $98 per cultivated acre with the yield effect, thus revealing yield effect costs of $55 per cultivated acre. These costs are as high as $77 per cultivated acre in the restoration

Table 3
Summary of wetland drainage scenarios considered in financial analysis.

Wetland Drainage Scenario	Description
Current Conditions	Regression model average yield response estimates are applied directly from the model, with unique values for each crop, associated with each impact code of wetland basin/buffer zone. The proportions of wetland basins adhering to each impact code in this scenario align with those described in Table 2.
Full Drainage	Under the full drainage scenario, all intact and partially drained wetlands are assumed to be drained. Therefore, the average yield response values for each crop in each location associated with intact and partially drained wetlands are changed to the average yield responses associated with drained wetlands and buffer zones. Farmed but undrained wetlands are assumed to remain undrained.
Restoration	Under the restoration scenario, all drained and partially drained wetlands are assumed to be intact. Therefore, the average yield response values for each crop in each location associated with drained or partially drained wetlands are changed to the average yield responses associated with intact wetlands and buffer zones.

Table 4
Annual per cultivated acre farm net benefits under three wetland drainage scenarios with sectional control technology and two yield response assumptions.

	Scenario		
	Full Drainage	Current Conditions	Restoration
Black Soil Zone			
% Cultivated Acres	100 %	96 %	93 %
% Wetlands Fully Drained/Farmed	100 %	58 %	30 %
No Yield Response Assumption			
Net benefits	$148	$153	$156
Difference in Profits Relative to Current Conditions	-$5	–	$3
Yield Response Assumption			
Net Benefits	$112	$98	$79
Difference in Profits Relative to Current Conditions	$14	–	-$19
Dark Brown Soil Zone			
% Cultivated Acres	100 %	97 %	95 %
% Wetlands Fully Drained/Farmed	100 %	59 %	18 %
No Yield Response Assumption			
Net Benefits	$181	$185	$189
Difference in Profits Relative to Current Conditions	-$4	–	$4
Yield Response Assumption			
Net Benefits	$173	$162	$156
Difference in Profits Relative to Current Conditions	$11	–	-$6

Table 5
Annual per cultivated acre farm net benefits under three wetland drainage scenarios, using average yield response indices under relatively wet and dry conditions, with sectional control technology, in the Black soil zone.

	Scenario		
Black Soil Zone	Full Drainage	Current Conditions	Restoration
Under Relatively Wet Conditions			
Net Benefits	$88	$74	$56
Difference in Profits Relative to Current Conditions	$14	–	-$18
Under Relatively Dry Conditions			
Net Benefits	$113	$98	$82
Difference in Profits Relative to Current Conditions	$15	–	-$16

Table 6
Annual per cultivated acre farm net benefits under three wetland drainage scenarios and two yield response assumptions, without the use of sectional control technology.

	Scenario		
	Full Drainage	Current Conditions	Restoration
Black Soil Zone			
No Yield Response Assumption, without Sectional Control			
Net Benefits	$148	$137	$128
Difference in Profits Relative to Current Conditions	$11	–	-$9
Yield Response Assumption, without Sectional Control			
Net Benefits	$112	$83	$51
Difference in Profits Relative to Current Conditions	$29	–	-$32
Dark Brown Soil Zone			
No Yield Response Assumption, without Sectional Control			
Net benefits	$181	$176	$170
Difference in Profits Relative to Current Conditions	$5	–	-$6
Yield Response Assumption, without Sectional Control			
Net Benefits	$173	$153	$138
Difference in Profits Relative to Current Conditions	$20	–	-$15

scenario, and $36 per cultivated acre in the full drainage scenario, exhibiting that wetland drainage mitigates these costs at the farm-level. This disparity implies that past estimates of wetland retention costs incurred by agricultural producers have been underestimated. The yield effect costs imposed by wetlands tend to be higher in the Black soil zone – indicative of spatial heterogeneity in the costs of wetland retention.

Although the consideration of wetland yield effects reduces per cultivated acre farm profitability, these effects increase wetland conservation costs, and as a result, the private incentives for wetland drainage. When yield effects are removed from the analysis, farmers are disincentivized from wetland drainage, as drainage costs exceed the benefits of nuisance/overlap cost reduction. This result aligns with neither the previous literature (though this could be a function of different cost structures and study timeframes), nor the observed trend of PPR wetland decline. When yield effects are included in the analysis, this trend reverses, and farmers are incentivized to drain wetlands on a per cultivated acre basis. While yields inside drained wetlands generally remain below the field average, the revenues from increases in productivity associated with drainage, including in wetland buffer zones, now exceed drainage costs. However, it should be emphasized that these effects are observed on a large and discrete scale, from full restoration to full drainage. The marginal incentives at the wetland-basin-level may not always reflect this broader trend, as the agronomic effects of individual wetland basins will ultimately determine the marginal benefits of drainage at smaller scales.

Sectional control technology alters the financial incentives for wetland drainage. When sectional control is utilized, the input overlap costs associated with farm wetland retention are reduced, decreasing the marginal benefits of wetland drainage. Sectional control could be considered a potential alternative to wetland drainage in some cases, particularly in fields with dispersed wetlands and poor navigability. However, the improved cost efficiency provided by sectional control is dwarfed in value by wetland yield effects. So, while widespread adoption of sectional control technology would reduce the private costs of wetlands, it cannot be considered a direct substitute for wetland drainage.

Although field edge effects have been studied in an agronomic context (Nguyen et al., 2022; Robinson et al., 2022), they had not been applied to economic analyses of farm wetland management to this point. We include wetland buffer zones into our analysis, and find that the agronomic effects of wetlands often extend more than 50 m beyond the wetland boundary. These effects are exacerbated by annual precipitation, increasing the incentives for wetland drainage in relatively wet years.

Our study complements and expands upon Clare et al. (2021) by incorporating data from more crops and field-years. Clare et al. (2021) found drained wetland basins yield at a profitable level on average, despite over half of individual basins yielding at a financial loss. Meanwhile, only 30 % of intact but farmed wetlands yielded at a financial loss. The authors suggest that this occurs because a large proportion of farmed wetland basins are temporary in nature, which would likely result in fewer agronomic issues relative to more permanent wetlands. Our data reflects this relationship – nearly 44 % of individual wetland basins in the sample are permanence class temporary and farmed through, though we find farmed and drained wetlands to yield at similar levels.[5] While our results are not directly comparable ($/wetland acre compared with $/cultivated acre, canola compared with a crop rotation, etc.), we find similar trends of broadly reduced productivity in wetland basins, but with enough variability in yield to suggest that individual wetlands can be highly productive under the right circumstances.

Since wetland retention presents clear opportunity and nuisance costs to landowners and farmers, payments to farmers can incentivize increased wetland conservation. Farmers often require high levels of compensation to participate in wetland conservation programs (Hill et al., 2011; Novak, 2017; Kauffman, 2018), and payments have often been insufficient to achieve certain conservation goals in the past (van Kooten and Schmitz, 1992; Cortus et al., 2011; Plews, 2017a, 2017b). The effects of wetland and buffer zone yield responses escalate the economic incentives for wetland drainage. This finding is vital from a policy perspective, and further suggests that wetland conservation payments may need to increase to properly compensate landowners for wetland retention.

Although this study presents important implications for wetland research and policy development moving forward, there are several limitations that should be noted. First, the study used observational data without experimental variation. The expression of yield as a percentage of the field average has certain benefits with respect to spatial and temporal comparability at the expense of observed yield variation within fields. Future studies can provide more robust estimates of wetland yield effects by incorporating and controlling for more factors, including input use, detailed soil and climate data, seeding and harvesting dates, pest pressure, and other wetland characteristics such as size and permanence. Input prices and quantities, crop yields, and crop prices were all derived from the 2022 Saskatchewan Crop Production Guide. Therefore, the results of the financial analysis may not align with actual financial outcomes of the fields and years in question. For example, if in a given year inputs are applied to achieve the 80th percentile target yield, but if weather is unfavourable, yields and profit margins would be reduced significantly. Changes in commodity prices have been shown to alter rates of drainage (De Laporte, 2014), so detailed input use and cost data at the sub-field level would improve the accuracy of the financial analysis.

The financial analysis is descriptive in nature and relies on assumptions which do not incorporate uncertainty. Therefore, the financial analysis results describe the annual net benefits of wetland drainage over the study time frame, but do not carry predictive interpretations, and therefore should only be considered as representative of the areas and years from which the yield data was collected. Incorporation of

[5] A full cross-section of wetland permanence class and impact code for the dataset and definitions of wetland permanence classes (Stewart and Kantrud, 1971) can be found in Appendix A.

variability in yield responses and associated impacts on agricultural producer decisions was beyond the purview of the financial analysis. The average yield response estimates are derived from multiple years of data, but are quantified in terms of a single growing season in the financial analysis. In reality, farmers are forward looking and will make drainage decisions based on several years of financial outcomes, and the financial incentives associated with these yield effect costs also will vary substantially over time. For example, while drainage is profitable at the assumed yields, prices, and crop input costs used in the financial analysis, variance in yields at the field and wetland-basin level can alter the financial incentives for drainage. The application of yield effects and the variability therein can produce stronger, more robust estimates of wetland conservation costs and incentives for drainage, and is an avenue for future research.

While a wetland basin's boundary and size are static and stable characteristics (van der Kamp et al., 2016), the area of standing water in the wetland will naturally fluctuate over time (Niemuth et al., 2010; van der Kamp et al., 2016; Zhang et al., 2021). These fluctuations are baked into the annual yield data but are not explicitly controlled for. Gauging the yield response to the extent of standing water in the sample wetlands would require more controlled observations of these fluctuations year-over-year. Finally, the assumption that wetland basins are static in drainage impact over time can be considered a potential source of measurement error in this analysis. Drainage activities may have occurred over the study period, and if so, may not have been captured in the data. Further, wetland basins classified as Code 2 (Farmed) wetlands tend to have lower degrees of permanence (see Appendix A). This implies that there may be temporal variability in the ability to consistently farm through these wetlands in wetter years which we are unable to account for. More precise measurement of these factors through time in future research can facilitate a stronger understanding of the relationships between wetlands and crop yields.

This research fills a key gap in understanding how wetlands and their buffer zones affect field crop yields and financial outcomes in the PPR. While opportunity and nuisance costs have been previously established as key drivers of farm wetland drainage, this research demonstrates the importance of considering the agronomic effects of wetlands and their buffer zones in future analyses. Increasingly available wetland mapping and precision agriculture data should facilitate a stronger understanding of these effects, increasing regulatory accuracy and effectiveness of wetland conservation efforts in the future.

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CRediT authorship contribution statement

Liam Boldt: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Patrick Lloyd-Smith:** Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Ken Belcher:** Writing – review & editing. **John Pattison-Williams:** Writing – review & editing. **Gary Bergen:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Katelyn Blechinger:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Ian Paulson:** Writing – review & editing, Methodology, Formal analysis, Data curation.

Declaration of competing interest

None.

Appendix A. Definitions of wetland permanence class and cross-tabulation counts and percentages of wetland permanence class and impact code in the study fields

Wetland class	Definition (Stewart and Kantrud, 1971)
1	Ephemeral – The wetland has free surface water for a short period of time.
2	Temporary – the wetland is periodically covered by water, typically lasting only a few weeks.
3	Seasonal ponds and lakes – the wetland is usually dry by midsummer.
4	Semi-permanent ponds and lakes – the wetland frequently maintains surface water throughout the growing season.
5	Permanent ponds and lakes – the wetland has permanent open water.

Wetland Characteristic		Impact Code				
		Intact	Partially Drained	Farmed	Drained	Total
Class	Temporary	158 (2.6 %)	35 (0.6 %)	2711 (43.9 %)	1055 (17.1 %)	3959 (64.1 %)
	Seasonal	465 (7.5 %)	121 (2.0 %)	466 (7.5 %)	596 (9.6 %)	1648 (26.7 %)
	Semi-permanent	166 (2.7 %)	97 (1.6 %)	49 (0.8 %)	95 (1.5 %)	407 (6.6 %)
	Permanent	86 (1.4 %)	27 (0.4 %)	5 (0.1 %)	47 (0.8 %)	165 (2.7 %)
	Total	875 (14.2 %)	280 (4.5 %)	3231 (52.3 %)	1793 (29.0 %)	6179 (100 %)

Appendix B. Mean and standard deviations of yields by crop and impact code through the 50 m buffer zone

Buffer Zone Increment (m)	0	5	10	15	20	25	30	35	40	45	50
Yield by… (St. Dev.) Crop											
Barley	80.72	86.06	89.94	91.86	92.33	92.55	93.45	93.96	94.72	95.60	96.02
	(43.3)	(36.6)	(33.2)	(30.5)	(28.7)	(28.1)	(26.6)	(25.0)	(23.9)	(23.4)	(22.9)
Canola	80.63	87.95	90.97	92.27	92.92	93.47	94.50	95.10	95.81	96.14	96.68
	(45.5)	(38.8)	(35.5)	(33.4)	(32.0)	(30.9)	(30.0)	(29.0)	(28.1)	(27.2)	(26.1)
Wheat	94.92	95.86	94.52	91.73	89.81	89.34	90.13	91.36	92.93	94.15	94.91
	(45.1)	(39.5)	(36.5)	(34.8)	(33.5)	(32.4)	(30.9)	(29.5)	(28.2)	(27.0)	(25.8)
Yellow Peas	64.92	75.81	81.35	82.94	84.17	85.96	87.88	88.81	90.20	91.25	92.53
	(42.1)	(40.1)	(39.2)	(37.2)	(36.2)	(35.4)	(34.0)	(32.0)	(30.3)	(29.6)	(28.5)

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Buffer Zone Increment (m)	0	5	10	15	20	25	30	35	40	45	50
Wetland Impact Code											
Drained	88.25	92.28	93.64	93.89	94.32	95.03	95.96	96.93	97.87	98.38	99.19
	(42.4)	(36.0)	(32.9)	(30.5)	(28.9)	(27.3)	(25.9)	(24.8)	(23.6)	(23.1)	(22.3)
Farmed	87.27	94.11	96.96	97.59	97.74	98.04	98.83	99.35	99.81	100.12	100.15
	(47.2)	(39.5)	(35.3)	(32.3)	(30.3)	(29.1)	(27.9)	(27.0)	(25.9)	(25.0)	(24.5)
Partially Drained	71.91	72.71	70.86	68.85	68.04	69.07	71.43	75.05	79.38	82.26	84.26
	(46.4)	(41.2)	(37.1)	(35.8)	(34.2)	(33.7)	(31.5)	(29.5)	(29.0)	(27.7)	(26.1)
Intact	74.27	77.52	78.27	76.94	75.66	75.85	77.72	79.03	81.27	83.29	85.02
	(44.4)	(39.9)	(37.5)	(35.8)	(35.0)	(34.1)	(33.3)	(31.5)	(30.7)	(30.0)	(28.2)
All Crops and Impact Codes	85.02	89.93	91.54	91.38	91.08	91.30	92.29	93.14	94.22	95.00	95.67
	(45.7)	(39.3)	(36.1)	(34.0)	(32.6)	(31.6)	(30.3)	(29.1)	(27.9)	(26.9)	(25.9)

Appendix C. Precipitation data used in the descriptive analysis

Precipitation data is derived from three weather stations in both the Black and Dark Brown soil zones, dispersed among the study fields (Government of Canada, 2022). The average of annual precipitation from the three weather stations in each soil zone is taken. Therefore, the analysis utilizes a separate value for average annual precipitation in the Black and Dark Brown soil zones in each year of data collection. Sites in the Black soil zone tend to receive more precipitation on average, and these differences in precipitation and soil type could drive regional differences in agricultural productivity in and around wetland areas.

C.1. Annual precipitation from each weather station in the Black and Dark Brown soil zones (mm/year)

Weather Station	2014	2015	2016	2017	2018	2019	2020
Black Soil Zone							
Elkhorn 2 East, MB	629	403	623	362	457	400	316
Virden, MB	–	588	338	246	365	459	263
Kipling, SK	604	429	591	273	452	477	280
Black Soil Zone Average	617	470	517	294	425	445	286
Dark Brown Soil Zone							
Last Mountain CS, SK	–	–	382	229	247	256	–
Moose Jaw CS, SK	–	–	515	218	229	405	–
Regina RCS, SK	–	–	437	152	204	375	–
Dark Brown Soil Zone Average	–	–	445	200	226	345	–

The annual precipitation data above is compared to the average annual precipitation for each soil zone to delineate wet and dry years in the descriptive analysis. If annual precipitation in a given year was greater than the average for that soil zone, it is classified as a wet year, and vice versa. Annual precipitation data by soil zone is obtained from McKenzie, 2019. Average annual precipitation is 437.5 mm in the Black soil zone and 350 mm in the Dark Brown soil zone.

Appendix D. Regression model parameter results

Black Soil Zone			
Location	Crop	Impact Code	Average Yield Response (Standard Error)
Wetland Basin	Barley	Drained	71.07*** (5.46)
		Farmed	75.94*** (7.02)
		Partially Drained	0.00
		Intact	0.00
	Canola	Drained	83.12*** (7.15)
		Farmed	71.47*** (8.88)
		Partially Drained	0.00
		Intact	0.00
	Wheat	Drained	91.93*** (8.67)
		Farmed	79.20*** (1.10)
		Partially Drained	0.00
		Intact	0.00
	Yellow Peas	Drained	50.47*** (2.68)
		Farmed	65.20*** (7.19)
		Partially Drained	0.00
		Intact	0.00
Buffer Zone	Barley	Drained	92.75*** (2.29)
		Farmed	95.45*** (1.28)
		Partially Drained	72.66*** (0.00)
		Intact	79.57*** (3.56)

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Black Soil Zone			
Location	Crop	Impact Code	Average Yield Response (Standard Error)
	Canola	Drained	97.48*** (2.74)
		Farmed	97.95*** (1.89)
		Partially Drained	82.44*** (3.69)
		Intact	77.92*** (4.25)
	Wheat	Drained	96.70*** (3.34)
		Farmed	97.96*** (1.86)
		Partially Drained	75.38*** (6.89)
		Intact	77.71*** (3.77)
	Yellow Peas	Drained	85.72*** (6.24)
		Farmed	94.95*** (5.01)
		Partially Drained	70.58*** (0.00)
		Intact	51.15*** (6.75)
Number of Observations	49,453		
Adjusted R^2	0.1893		

Dark Brown Soil Zone			
Location	Crop	Impact Code	Average Yield Response (Standard Error)
Wetland Basin	Canola	Drained	106.28*** (4.69)
		Farmed	115.38*** (12.27)
		Partially Drained	0.00
		Intact	0.00
	Wheat	Drained	107.41*** (6.52)
		Farmed	114.51*** (12.61)
		Partially Drained	0.00
		Intact	0.00
	Yellow Peas	Drained	68.10** (12.01)
		Farmed	68.37*** (9.41)
		Partially Drained	0.00
		Intact	0.00
Buffer Zone	Canola	Drained	102.05*** (1.74)
		Farmed	103.96*** (3.18)
		Partially Drained	77.47*** (6.72)
		Intact	82.08*** (2.18)
	Wheat	Drained	93.88*** (1.82)
		Farmed	98.53*** (3.33)
		Partially Drained	70.47*** (0.73)
		Intact	82.93*** (1.32)
	Yellow Peas	Drained	89.05*** (3.38)
		Farmed	95.28*** (2.83)
		Partially Drained	57.26*** (6.41)
		Intact	79.64*** (4.79)
Number of Observations	27,669		
Adjusted R^2	0.1808		

Notes: Statistical significance indicators: * Significance at $p < 0.10$; ** Significance at $p < 0.05$; *** Significance at $p < 0.01$.

Appendix E. Summary of costs and revenues from field crop production used in the financial analysis (Government of Saskatchewan, 2022)

Black Soil Zone	Crop			
Field Crop Revenues	Wheat	Canola	Peas	Barley
Target Yield (bu/ac)	64.3	51.15	55.12	73.73
Price ($/bu)	$10.56	$17.01	$12.00	$6.11
Baseline Revenue ($/ac)	$679.01	$870.06	$661.44	$450.49
Input Costs ($/ac)				
Seed	$30.60	$75.73	$71.20	$40.61
Seed Treatments/Inoculants	$0.84	$9.00	$12.31	$1.02
Fertilizer				
Nitrogen	$141.15	$143.81	$11.85	$103.86
Phosphorous (P2O5)	$35.81	$49.45	$35.81	$28.99
Sulphur and Other	$0.00	$9.21	$0.00	$0.00
Crop Protection				
Herbicides	$63.33	$66.28	$72.41	$63.78
Insecticides	$21.89	$2.46	$15.22	$21.89
Fungicides	$19.35	$14.18	$14.18	$19.35
Machinery Operating				
Fuel	$19.14	$20.27	$21.39	$19.14
Repair	$11.29	$11.29	$11.29	$11.29
Custom Work and Hired Labour	$23.05	$21.05	$20.30	$21.05

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Black Soil Zone	Crop			
Field Crop Revenues	Wheat	Canola	Peas	Barley
Crop Insurance Premium	$4.78	$10.96	$6.01	$4.68
Hail Insurance Premium	$12.25	$12.25	$12.25	$12.25
Utilities and Miscellaneous	$4.88	$4.88	$4.88	$4.88
Interest on Variable Expenses	$7.79	$9.05	$6.20	$7.08
Other (buildings, property, machinery)	$116.87	$116.87	$116.87	$116.87
Total Cost ($/ac)	$513.02	$576.74	$432.18	$476.75
Net Profit ($/ac)	$165.99	$293.32	$229.26	$(26.26)

Dark Brown Soil Zone	Crop		
Field Crop Revenue	Wheat	Canola	Peas
Target Yield (bu/ac)	57.32	47.18	48.13
Price ($/bu)	$10.56	$17.01	$12.00
Baseline Revenue ($/ac)	$ 605.30	$ 802.53	$ 577.56
Input Costs ($/ac)			
Seed	$26.92	$75.73	$63.20
Seed Treatments/Inoculants	$0.74	$9.00	$10.93
Fertilizer			
Nitrogen	$126.50	$133.16	$10.39
Phosphorous (P2O5)	$31.55	$46.04	$31.55
Sulphur and Other	$0.00	$8.42	$0.00
Crop Protection			
Herbicides	$59.95	$58.24	$66.08
Insecticides	$21.89	$2.46	$15.22
Fungicides	$19.35	$14.18	$14.18
Machinery Operating			
Fuel	$15.31	$16.21	$17.12
Repair	$9.98	$9.98	$9.98
Custom Work and Hired Labour	$22.05	$21.05	$20.30
Crop Insurance Premium	$4.59	$10.51	$5.14
Hail Insurance Premium	$12.25	$12.25	$12.25
Utilities and Miscellaneous	$4.23	$4.23	$4.23
Interest on Variable Expenses	$7.13	$8.46	$5.63
Other (buildings, property, machinery)	$107.53	$107.53	$107.53
Total Cost ($/ac)	$469.96	$537.45	$393.71
Net Profit ($/ac)	$135.34	$265.09	$183.85

Appendix F. Agronomic assumptions from the Black and Dark Brown soil zones (Government of Saskatchewan, 2022)

Crop	Seed	Fertilizer	Plant Protection	Machinery Operating	Hired Labour	Crop Insurance Premium	Utilities and Misc.	Interest Expenses	Other
Black Soil Zone									
Wheat	25 plants/ sq. ft. target plant stand.	Nitrogen: 78 lb./ac P2O5: 34 lb./ac.	Based on provincial insect, disease, and weed pressure.	Based on diesel priced $0.901/ l with a repair rate of 2.6 % of yearly machinery investment.	Labour assumed to be $26.40 per hour for custom farm operations.	Five-year averages of premiums by producers who attained targeted yield.	Cost of electricity, natural gas, water, and telephone expenses.	A rate of 3.01 % used on all variable expenses applied for 8 months.	Buildings, property, and machinery.
Canola	5 lb./ac. seeding rate.	Nitrogen: 108 lb./ac. P2O5: 58 lb./ac. Sulphur: 15 lb./ac.							
Yellow Peas	178 lb./ac. seeding rate.	Nitrogen: 9 lb./ac. P2O5: 42 lb./ac.							
Barley	25 plants/ sq. ft. target plant stand.	Nitrogen: 78 lb./ac. P2O5: 34 lb./ac.							
Dark Brown Soil Zone									
Wheat	22 plants/ sq. ft. target plant stand	Nitrogen: 95 lb./ac P2O5: 37 lb./ac.	Based on provincial insect, disease, and weed pressure.	Based on diesel priced $0.901/ l with a repair rate of 2.6 % of yearly machinery investment.	Labour assumed to be $26.40 per hour for custom farm operations.	Five-year averages of premiums by producers who attained targeted yield.	Cost of electricity, natural gas, water, and telephone expenses.	A rate of 3.01 % used on all variable expenses applied for 8 months.	Buildings, property, and machinery.

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Crop	Seed	Fertilizer	Plant Protection	Machinery Operating	Hired Labour	Crop Insurance Premium	Utilities and Misc.	Interest Expenses	Other
Canola	5 lb./ac. seeding rate.	Nitrogen: 100 lb./ac. P2O5: 54 lb./ac. Sulphur: 15 lb./ac.							
Yellow Peas	158 lb./ac. seeding rate.	Nitrogen: 8 lb./ac. P2O5: 37 lb./ac.							

Appendix G. Wetland and corrected buffer zone acres by impact code and soil zone

		Black Soil Zone	Dark Brown Soil Zone
Impact Code		Wetland Area (acres)	
Intact	(0)	117	152
Partially Drained	(1)	69	111
Farmed	(2)	113	117
Drained	(5)	127	256
Total		446	637
Impact Code		Corrected Buffer Area (acres)	
Intact	(0)	558	888
Partially Drained	(1)	215	387
Farmed	(2)	1369	1844
Drained	(5)	731	1367
Total		2873	4486

Appendix H. Yield response matrices for each soil zone and drainage scenario

H.1. Yield response matrix in the Black soil zone, restoration scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)			
			Wheat	Canola	Yellow Peas	Malt Barley
0	Wetland	0	0	0	0	0
1	Wetland	0	0	0	0	0
2	Wetland	2	0.792	0.715	0.652	0.759
5	Wetland	0	0	0	0	0
0	5-50 m Buffer	0	0.777	0.779	0.511	0.796
1	5-50 m Buffer	0	0.777	0.779	0.511	0.796
2	5-50 m Buffer	2	0.98	0.98	0.949	0.955
5	5-50 m Buffer	0	0.777	0.779	0.511	0.796

H.2. Yield response matrix in the Black soil zone, current conditions scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)			
			Wheat	Canola	Yellow Peas	Malt Barley
0	Wetland	0	0	0	0	0
1	Wetland	1	0	0	0	0
2	Wetland	2	0.792	0.715	0.652	0.759
5	Wetland	5	0.919	0.831	0.505	0.711
0	5-50 m Buffer	0	0.777	0.779	0.511	0.796
1	5-50 m Buffer	1	0.754	0.824	0.706	0.727
2	5-50 m Buffer	2	0.980	0.980	0.949	0.955
5	5-50 m Buffer	5	0.967	0.975	0.857	0.928

H.3. Yield response matrix in the Black soil zone, full drainage scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)			
			Wheat	Canola	Yellow Peas	Malt Barley
0	Wetland	5	0.919	0.831	0.505	0.711
1	Wetland	5	0.919	0.831	0.505	0.711
2	Wetland	2	0.792	0.715	0.652	0.759
5	Wetland	5	0.919	0.831	0.505	0.711
0	5-50 m Buffer	5	0.967	0.975	0.857	0.928
1	5-50 m Buffer	5	0.967	0.975	0.857	0.928
2	5-50 m Buffer	2	0.980	0.980	0.949	0.955
5	5-50 m Buffer	5	0.967	0.975	0.857	0.928

H.4. Yield response matrix in the Dark Brown soil zone, restoration scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Yellow Peas
0	Wetland	0	0	0	0
1	Wetland	0	0	0	0
2	Wetland	2	1.145	1.154	0.684
5	Wetland	0	0	0	0
0	5-50 m Buffer	0	0.829	0.821	0.796
1	5-50 m Buffer	0	0.829	0.821	0.796
2	5-50 m Buffer	2	0.985	1.040	0.953
5	5-50 m Buffer	0	0.829	0.821	0.796

H.5. Yield response matrix in the Dark Brown soil zone, current conditions scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Yellow Peas
0	Wetland	0	0	0	0
1	Wetland	1	0	0	0
2	Wetland	2	1.145	1.154	0.684
5	Wetland	5	1.074	1.063	0.681
0	5-50 m Buffer	0	0.829	0.821	0.796
1	5-50 m Buffer	1	0.705	0.775	0.573
2	5-50 m Buffer	2	0.985	1.040	0.953
5	5-50 m Buffer	5	0.939	1.020	0.890

H.6. Yield response matrix in the Dark Brown soil zone, full drainage scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Yellow Peas
0	Wetland	5	1.074	1.063	0.681
1	Wetland	5	1.074	1.063	0.681
2	Wetland	2	1.145	1.154	0.684
5	Wetland	5	1.074	1.063	0.681
0	5-50 m Buffer	5	0.939	1.020	0.890
1	5-50 m Buffer	5	0.939	1.020	0.890
2	5-50 m Buffer	2	0.985	1.040	0.953
5	5-50 m Buffer	5	0.939	1.020	0.890

H.7. Yield response matrix under wet conditions in the Black soil zone, restoration scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Malt Barley
0	Wetland	0	0	0	0
1	Wetland	0	0	0	0
2	Wetland	2	0.871	0.648	0.687
5	Wetland	0	0	0	0
0	5-50 m Buffer	0	0.778	0.646	0.769
1	5-50 m Buffer	0	0.778	0.646	0.769
2	5-50 m Buffer	2	0.981	0.945	0.941
5	5-50 m Buffer	0	0.778	0.646	0.769

H.8. Yield response matrix under wet conditions in the Black soil zone, current conditions scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Malt Barley
0	Wetland	0	0	0	0
1	Wetland	1	0	0	0
2	Wetland	2	0.871	0.648	0.687
5	Wetland	5	0.920	0.724	0.655
0	5-50 m Buffer	0	0.778	0.646	0.769
1	5-50 m Buffer	1	0.720	0.735	0.817
2	5-50 m Buffer	2	0.981	0.945	0.941
5	5-50 m Buffer	5	0.953	0.926	0.896

H.9. Yield response matrix under wet conditions in the Black soil zone, full drainage scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Malt Barley
0	Wetland	5	0.920	0.724	0.655
1	Wetland	5	0.920	0.724	0.655
2	Wetland	2	0.871	0.648	0.687
5	Wetland	5	0.920	0.724	0.655
0	5-50 m Buffer	5	0.953	0.926	0.896
1	5-50 m Buffer	5	0.953	0.926	0.896
2	5-50 m Buffer	2	0.981	0.945	0.941
5	5-50 m Buffer	5	0.953	0.926	0.896

H.10. Yield response matrix under dry conditions in the Black soil zone, restoration scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Malt Barley
0	Wetland	0	0	0	0
1	Wetland	0	0	0	0
2	Wetland	2	0.704	0.750	0.821
5	Wetland	0	0	0	0
0	5-50 m Buffer	0	0.773	0.817	0.826
1	5-50 m Buffer	0	0.773	0.817	0.826
2	5-50 m Buffer	2	0.978	1.000	0.969
5	5-50 m Buffer	0	0.773	0.817	0.826

H.11. Yield response matrix under dry conditions in the Black soil zone, current conditions scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Malt Barley
0	Wetland	0	0	0	0
1	Wetland	1	0	0	0
2	Wetland	2	0.704	0.750	0.821
5	Wetland	5	0.918	0.882	0.794
0	5-50 m Buffer	0	0.779	0.817	0.826
1	5-50 m Buffer	1	0.826	0.847	0.677
2	5-50 m Buffer	2	0.978	1.000	0.969
5	5-50 m Buffer	5	1.005	1.000	0.980

H.12. Yield response matrix under dry conditions in the Black soil zone, full drainage scenario

Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Malt Barley
0	Wetland	5	0.918	0.882	0.797
1	Wetland	5	0.918	0.882	0.797

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Original Data Impact Code	Polygon Type	Scenario Impact Code	Yield Response Index (yield as % of field average)		
			Wheat	Canola	Malt Barley
2	Wetland	2	0.704	0.750	0.821
5	Wetland	5	0.918	0.882	0.797
0	5-50 m Buffer	5	1.005	1.000	0.980
1	5-50 m Buffer	5	1.005	1.000	0.980
2	5-50 m Buffer	2	0.978	1.000	0.969
5	5-50 m Buffer	5	1.005	1.000	0.980

Data availability

The data that has been used is confidential.

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